

ENRO-VET 200 mg/ml ORAL SOLUTION

FOR ANIMAL USE ONLY

PRODUCT NAME:

ENRO-VET 200 mg/ml ORAL SOLUTION

NAME AND STRENGTH OF ACTIVE INGREDIENT:

Each ml contains 200 mg Enrofloxacin

PRODUCT DESCRIPTION

A clear and yellowish solution.

PHARMACODYNAMIC PROPERTIES:

Mode of action

Two enzymes essential in DNA replication and transcription, DNA gyrase and topoisomerase IV, have been identified as the molecular targets of fluoroquinolones. They modulate the topological state of DNA through cleaving and resealing reactions. Initially, both strands of the DNA double helix are cleaved. Then, a distant segment of DNA is passed through this break before the strands are resealed. Target inhibition is caused by non-covalent binding of fluoroquinolone molecules to an intermediate state in this sequence of reactions, in which DNA is cleaved, but both strands are retained covalently attached to the enzymes. Replication forks and translational complexes cannot proceed beyond such enzyme-DNA-fluoroquinolone complexes, and inhibition of DNA and mRNA synthesis triggers events resulting in a rapid, drug concentration-dependant killing of pathogenic bacteria.

Antibacterial spectrum

Enrofloxacin is active against many Gram-negative bacteria, against Gram-positive bacteria and *Mycoplasma* spp.

In vitro susceptibility has been shown in strains of (i) Gram-negative species such as, *Pasteurella multocida* and *Avibacterium (Haemophilus) paragallinarum* and (ii) *Mycoplasma gallisepticum* and *Mycoplasma synoviae*.

Types and mechanisms of resistance

Resistance to fluoroquinolones has been reported to arise from five sources, (i) point mutations in the genes encoding for DNA gyrase and/or topoisomerase IV leading to alterations of the respective enzyme, (ii) alterations of drug permeability in Gram-negative bacteria, (iii) efflux mechanisms, (iv) plasmid mediated resistance and (v) gyrase protecting proteins. All mechanisms lead to a reduced susceptibility of the bacteria to fluoroquinolones. Cross-resistance within the fluoroquinolone class of antimicrobials is common.

PHARMACOKINETIC PROPERTIES:

Enrofloxacin has a relatively high oral bioavailability in almost all the studied species. Concomitant administration of compounds containing multivalent cations (antacids, milk, or milk substitutes) decreases the oral bioavailability of fluoroquinolones. Fluoroquinolones are characterized by an extensive diffusion to body fluids and tissues, reaching in some concentrations higher than those found in plasma. Furthermore, they are widely distributed in skin, bone and semen, also reaching the anterior and posterior chambers of the eye; they cross the placenta and the blood-brain barrier. They are also stored in phagocytic cells (alveolar macrophages, neutrophils) and are therefore effective against intracellular microorganisms. The biotransformation of enrofloxacin at the hepatic level gives rise to an active metabolite which is ciprofloxacin. In general, metabolism occurs through processes of hydroxylation and oxidation to oxofluoroquinolones. Other reactions that also occur are N-dealkylation and glucuronic acid conjugation. Excretion occurs through the biliary and renal routes, the latter being the predominant one. Renal excretion is carried out by glomerular filtration, and also by active tubular secretion through the organic anion pump.

INDICATIONS:

Treatment of infections caused by the following enrofloxacin sensitive bacteria:

Chickens

Mycoplasma gallisepticum,
Mycoplasma synoviae,
Avibacterium paragallinarum,
Pasteurella multocida,

TARGET SPECIES

Poultry (chickens)

ROUTE OF ADMINISTRATION

Solution for administration in drinking water

DOSAGE

Chickens:

10 mg enrofloxacin / kg bw / day, for 3-5 consecutive days (equivalent to 0.05 ml of medicine / kg bw / day).

Treatment for 5 consecutive days in case of mixed infections and in chronic processes. If no clinical improvement is seen within 2-3 days, treatment should be reconsidered based on sensitivity tests.

Animals should be weighed as accurately as possible to avoid insufficient dosing.

Water consumption depends on the animal's clinical situation and the time of year.

To ensure a correct dosage, the concentration of enrofloxacin in the water will be adjusted taking into account daily consumption.

Based on the recommended dose, number, and weight of animals to be treated, the exact daily dose of medication should be calculated using the following formula:

$$\text{ml of medicine / L of water} = \frac{10 \text{ mg / kg / day} \times \text{mean weight of animals}}{200 \text{ mg / ml} \times \text{average water consumption (L / day)}}$$

Direction of use:

Medicated water should be prepared every 24 hours, immediately before starting treatment. Medicated water should be the only source of water throughout the treatment.

Check that the administered dose is completely consumed by checking the delivery systems periodically. Before starting the treatment, empty the supply system and fill it with medicated water.

Use appropriate and properly calibrated dosing equipment.

CONTRAINDICATIONS

Do not use for prophylaxis.

Do not use if resistance or cross-resistance to (fluoro) quinolones has been identified in the group of birds to be treated.

Do not use in cases of known hypersensitivity to the active substance, other (fluoro) quinolones or any excipients.

SPECIAL WARNINGS FOR EACH TARGET SPECIES

Treatment of *Mycoplasma* spp. infections may not eradicate microorganism.

SPECIAL PRECAUTION FOR USE

Special precautions for use in animals

Official and local antimicrobial policies should be taken into account when the product is used. Fluoroquinolones should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly, to other classes of antimicrobials. Since enrofloxacin was first authorised for use in poultry, there has been widespread reduction in susceptibility of *E. coli* to fluoroquinolones and emergence of resistant organisms. Resistance has also been reported in *Mycoplasma synoviae* in the EU.

Whenever possible, fluoroquinolones should only be used based on susceptibility testing. Use of the product deviating from the instructions given in the label may increase the prevalence of bacteria resistant to the fluoroquinolones and may decrease the effectiveness of treatment with other quinolones due to the potential for cross resistance.

Special precautions to be taken by the person administering the medicinal product to animals

Those with known hypersensitivity to (fluoro)quinolones should avoid contact with this product. Avoid contact with skin and eyes. Rinse any splashes from skin or eyes immediately with water. Wash hands and exposed skin after use. Do not eat, drink or smoke while using the product.

INTERACTIONS WITH OTHER MEDICAMENTS

In vitro, an antagonistic effect has been demonstrated when fluoroquinolones are combined with other bacteriostatic antimicrobials such as macrolides or tetracyclines and phenols. Concomitant administration of substances containing magnesium or aluminum may reduce the absorption of enrofloxacin.

USE DURING PREGNANCY, LACTATION AND LAY

Laboratory studies in rats have not shown teratogenic effects. Studies in rabbits have not shown teratogenic effects for the fetus or the mother.

Studies in rabbits have shown no toxic effects for lactating rabbits during the first 16 days of life. Rabbits older than this age have the ability to eliminate enrofloxacin.

Use only in accordance with the benefit / risk assessment by the responsible veterinarian.

Its use is not authorized in laying birds whose eggs are used for human consumption.

Do not use in replacement laying birds in the 14 days prior to the start of lay.

ADVERSE EFFECTS

None known

OVERDOSE AND TREATMENT

In the event of overdose, the symptoms would consist of a weak stimulation of spontaneous motility, so treatment should be discontinued.

Fluoroquinolone poisoning can cause nausea, vomiting, and diarrhea.

WITHDRAWAL PERIODS

Chickens (meat): 7 days.

Its use is not authorized in laying birds whose eggs are used for human consumption.

Do not use in replacement laying birds in the 14 days prior to the start of lay.

SHELF-LIFE

Shelf life as packaged for sale: 2 year

Shelf life after first opening : 24 hours

Shelf life after reconstitution : 24 hours

ALLOWABLE MAXIMUM RESIDUAL LIMIT (MRL)

Sum of enrofloxacin and ciprofloxacin: Muscle, liver, kidney (chicken) = 30 µg/kg (Reference: REGOVP Third Version, July 2014)

STORAGE CONDITIONS

Store in a cool place below 30°C. Keep out of reach of children. Jauhi ubat daripada kanak-kanak.

DISPOSAL OF CONTAINERS

Any unused veterinary medicinal product or waste material derived from such veterinary medicinal products should be disposed of in accordance with local requirements.

PACKAGING AVAILABLE

Available in 500ml, 1L HDPE bottle.

MANUFACTURER AND PRODUCT REGISTRATION HOLDER

Life Biopharma Sdn. Bhd. (company no. 546163-D)

No. 250-251, Jalan Industri Galla 13, Kawasan Perindustrian Galla, 70200 Seremban, Negeri Sembilan, Malaysia.

DATE OF REVISION OF PACKAGE INSERT

19th October 2020